

Aspen Aerogels, Inc. NYSE:ASPN

FQ3 2023 Earnings Call Transcripts

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S&P Global Market Intelligence Estimates

	-FQ3 2023-			-FQ4 2023-	-FY 2023-	-FY 2024-
	CONSENSUS	ACTUAL	SURPRISE	CONSENSUS	CONSENSUS	CONSENSUS
EPS Normalized	(0.27)	(0.19)	NM	(0.22)	(0.96)	(0.37)
Revenue (mm)	53.30	60.76	▲ 14.00	80.94	228.86	406.13

Currency: USD

Consensus as of Nov-02-2023 10:10 AM GMT

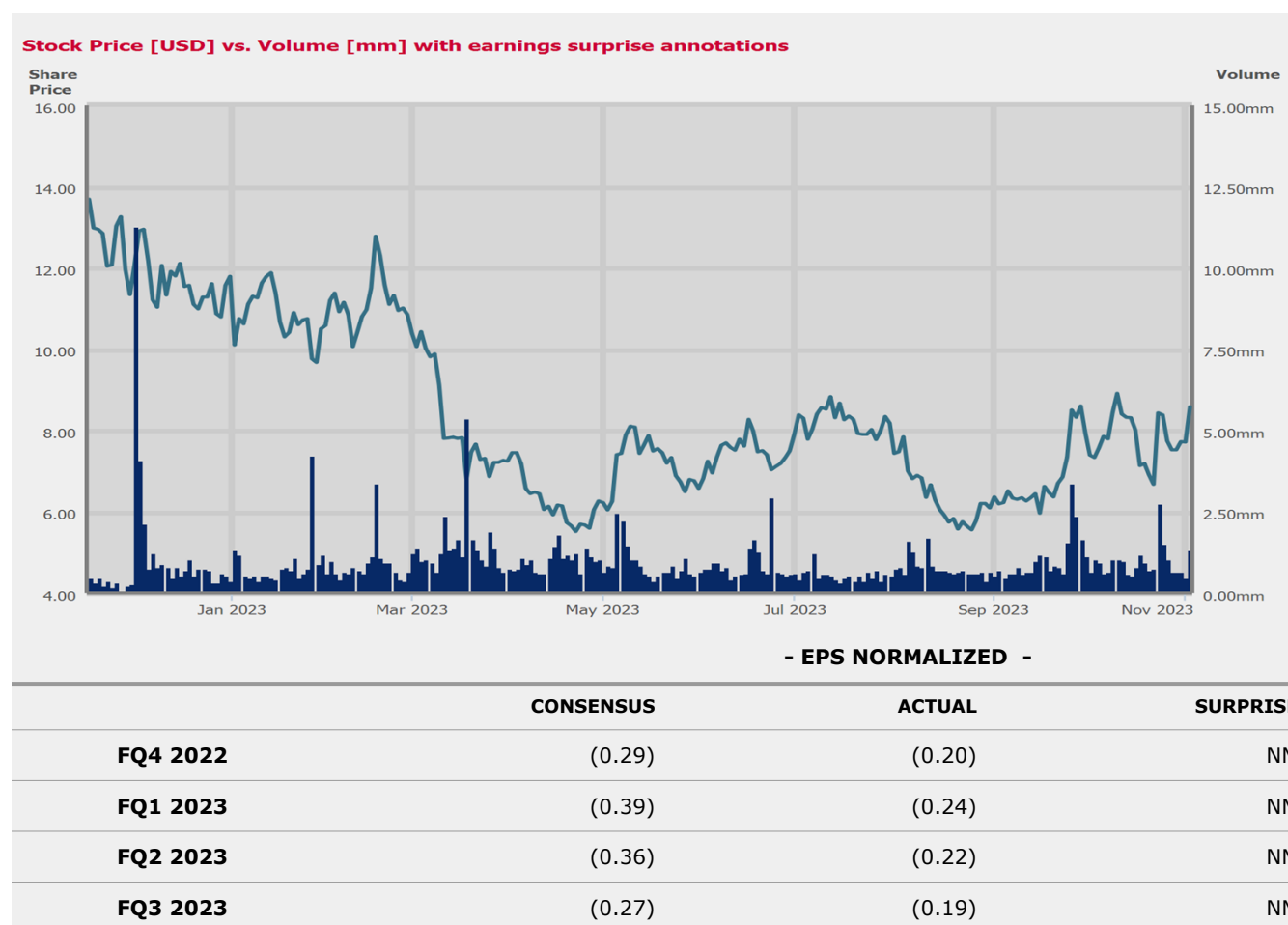


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Call Participants

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Presentation

Operator

Good morning. Thank you for attending Aspen Aerogels, Inc. Q3 2023 Financial Results Call. [Operator Instructions]

I would now like to turn the conference over to your host, Neal Baranosky, Aspen's Senior Director of Corporate Strategy and Finance. Thank you. You may proceed, Mr. Baranosky.

Neal Baranosky

Thank you, Elliot. Good morning, and thank you for joining us for the Aspen Aerogels' Fiscal Year 2023 Third Quarter Financial Results Conference Call. With us today are Don Young, President and CEO; and Ricardo Rodriguez, Chief Financial Officer.

There are a few housekeeping items that I would like to address before turning the call over to Don. The press release announcing Aspen's financial results and business developments as well as a reconciliation of management's use of non-GAAP financial measures compared to the most applicable U.S. generally accepted accounting principles, or GAAP measures, is available on the Investors section of Aspen's website, www.aerogel.com.

In addition, I'd like to highlight that we've uploaded to our website a slide deck that will accompany our conversation today. You can find the deck in the Investors section on our website.

On today's call, management will make forward-looking statements about our expectations. These statements are subject to risks and uncertainties that could cause our actual results to differ materially. These risks and uncertainties include the factors identified in our filings with the SEC. Please review the disclaimer statements on Pages 1 and 2 of the slide deck, as the content of our call will be governed by this language.

During this call, we will refer to non-GAAP financial measures, including adjusted EBITDA. These financial measures are not prepared in accordance with GAAP. These non-GAAP financial measures are not intended to be considered in isolation or as a substitute for results prepared in accordance with GAAP.

The definitions and reconciliations of these non-GAAP financial measures to the most directly comparable GAAP financial measures and a discussion of why we present these non-GAAP financial measures are included in yesterday's press release.

And one final note, during the Q&A session, in the interest of time, we ask that you limit your questions to 2 questions at a time. If you have additional questions beyond the initial 2, please get back into the queue and we will get to all questions.

I'll now turn the call over to Don. Don?

Donald R. Young

President, CEO & Director

Thanks, Neal. Good morning, everyone. Thank you for joining us for our Q3 2023 earnings call. My initial comments will highlight our Q3 performance and Q4 outlook, the status and impact of several critical elements of our strategy and our EV OEM development pipeline. Ricardo will dig deeper into our financial performance and our business strategy. Ricardo and I will expand upon last week's announcement related to our upgraded revenue and adjusted EBITDA outlook for 2023 and the addition of Audi and Scania to our list of OEM design awards. We will conclude with a Q&A session.

Q3 revenue was a record at over \$60 million. PyroThin Thermal Barrier revenue was \$33 million, surpassing that of our Energy Industrial business for the first time. We believe PyroThin Thermal Barrier revenue for the second half of 2023 will be 3 or [4 times] larger than it was for the first half of the year. Gross profit for the second half of 2023 will be 2 or [3 times] larger than for the first half of the year.

Energy Industrial activity remained strong, while the team manages through a supply constrained period as we shift Plant 1 to produce PyroThin Thermal Barriers and [co-test] the system of our plant's supplemental supply for Energy Industrial. We are making good progress on both initiatives.

In addition to the ramping of the PyroThin Thermal Barrier business, a highlight for Q3 is the continued progression of our gross margins through 2023, 11% in Q1, 17% in Q2, 23% in Q3. We anticipate continued gross margin expansion in Q4 as we progress on the path to achieving our targeted 35% gross margin.

Based on our Q3 performance and current momentum, on October 24 we announced a revenue outlook of at least \$225 million, favorably modified from an earlier outlook of a range between \$200 million and \$250 million. The UAW strike has not impacted our revenue, but we will remain cautious until the tentative agreements have been ratified, and the auto workers are fully back to work.

We also announced an improvement to our 2023 adjusted EBITDA outlook from a midpoint of negative \$50 million to a midpoint of negative \$35 million, our second enhanced adjusted EBITDA outlook in as many quarters. The improved outlook for adjusted EBITDA is driven by higher volumes, leading to a fuller fixed cost absorption. We expect this trend to continue in Q4.

Also on October 24, we announced the additions of Scania and Audi, both part of the Volkswagen Group, to our list of design awards. We expect revenue from these 2 awards to commence during 2024, scale in 2025 and be significant contributors in 2026 and beyond. We anticipate the opportunity to serve other European OEMs as well, and continue to believe that we remain on track to reach our target of 6 OEM design awards by year-end.

2 key elements of the implementation of our strategy are: first, the full conversion of Plant 1 in East Providence to PyroThin Thermal Barrier supply; and second, the commencement of our supplemental supply dedicated to our Energy Industrial segment. We believe that both strategic initiatives will be functional in early 2024. A third key element of our strategy is the balancing of growth, scale and profitability, which includes the right timing of Plant 2 in Georgia.

Our focus prior to the restart of full construction of Plant 2 is to utilize our existing assets, supply arrangements and current commercial opportunities to build a business that has the potential to produce annually approximately \$550 million of revenue, approximately \$200 million of gross profit, and approximately \$140 million of EBITDA. We believe we are well positioned to attain this level of performance.

Current industry headlines citing uncertainty related to EV capacity investments by OEMs and to the ramp for EV demand are consistent with the underlying assumptions we have used since early 2023 in forming our operating and investment plans. Given the challenging capital environment, we are working to avoid unnecessary dilution for our shareholders by seeking to operate from a position of operational and financial strength, and potentially by partnering with the DOE loan programs office with whom we remain in close contact regarding our advanced technology vehicle manufacturing loan application.

We believe a more measured ramp in OEM EV production may enable us to have a capital-efficient growth path utilizing our current assets, that leads to profitability in the near term and without sacrificing our full growth opportunities in the longer term.

Ricardo, over to you.

Ricardo C. Rodriguez
CFO & Treasurer

Thank you, Don, and good morning, everyone. I'll start by covering our third quarter and year-to-date results before moving on to the 2023 outlook and then handing the call back to Don.

For this quarterly call I don't have any meaningful strategic updates to cover, and I'm happy to simply focus on reporting our results, with the third quarter representing an important transition period towards higher revenue run-rate levels that enable meaningful reductions in our operating losses. Our strategy is

also yielding positive developments in our commercial pipeline, such as the recent conversion of the LOI from Audi into an award, and are gearing up of production to begin supplying Scania at higher volumes next year.

As Don mentioned, we feel confident in our ability to receive additional OEM awards as we focus on delivering results with our existing assets.

To cover our performance, I'll start on Slide 4, beginning with revenues. We delivered \$60.8 million of revenue in Q3, which translates into 66% growth year-over-year, and 26% growth quarter-over-quarter. As Don mentioned, this was an all-time company record. We didn't suffer any supply disruptions, thanks to the team's efforts over the past 12 months in Rhode Island and Mexico to preemptively manage our supply chain. Year-to-date we have delivered \$154.5 million of revenue, which reflects a 28% year-over-year increase. Year-to-date Energy Industrial revenues were \$97.3 million, an 8% year-over-year increase.

Given our limited area of production capacity, in Q3, we continued to focus on optimizing our Energy Industrial product mix to lighten the load on our operations by making those products that require the least standard hours of processing and deliver \$27.9 million in sales, reflecting a 21% quarterly decrease and a 13% year-over-year increase.

As we previously mentioned, our energy business is sold out. We have a clear line of sight to about \$216 million of annual demand and currently have approximately \$118 million of backlog in orders to fulfill over the next few quarters. To fulfill this excess demand, we are focused on continuing to optimize our mix for steady supply during the remainder of the year, and bringing in supplemental supply as soon as possible.

EV Thermal Barrier revenues of \$32.8 million were up 175% year-over-year and 160% quarter-over-quarter, reflecting the accelerating ramp in GM's production of Ultium platform-based electric vehicles, and steady volumes in the Toyota nameplate that we supply, the bZ4X. General Motors has more than doubled the production of models that we supply them on quarter-over-quarter. Our year-to-date EV Thermal Barrier revenues were \$57.2 million, representing an 88% increase when compared to the same period in 2022.

Next, I'll provide a summary of our main expenses. Material expenses of \$22 million for the quarter made up 36 percentage points of sales, continuing to reflect the work that our supply chain and procurement groups have put in to reducing the cost of some of our main raw materials. We see this as temporary relief and the result of conservative planning as well. So, we will remain vigilant with the goal of ensuring that we can keep these below 40 percentage points of sales. The Q3 performance enabled our total year-to-date material cost to be of \$58 million or 38 percentage points of sales, or 200 basis points favorable to our target of 40 percentage points of sales.

Conversion costs, which we describe as all production costs required to convert raw materials into finished products were of \$25 million or 41 percentage points of sales in Q3. These costs include all elements of the direct labor, manufacturing, overhead, factory supplies, brand, insurance, process, logistics, quality and inspection. These results compare favorably to conversion costs in Q2 of this year, which were a 46 percentage points of sales. This is the result of much better fixed cost absorption on our aerogel production costs driven by the higher sales run rate level of this quarter.

As previously mentioned, our long-term target for these costs at a roughly double revenue run rate, it's a 20 to 25 percentage points of sales. So, we aren't done managing these. While we've made improvements, primarily thanks to the efficiency of our operations in Mexico, we need to continue capturing additional opportunities to reduce these costs. Looking ahead, as our teams start running all lines of our aerogel plant in Rhode Island to make PyroThin without any switchovers and find their flow, we're sure that we'll see some efficiencies.

In Mexico, for EV Thermal Barrier part assembly, this week we have a focused team with all hands on deck on increasing the uptime and throughput of our encapsulation and assembly equipment to drive efficiency and enable the next potential step up in demand to launch next year of simpler designs for prismatic

cells on a new set of encapsulation equipment that can be shared across programs will drive additional efficiency.

Year-to-date, our conversion costs of \$69.2 million reflect 45 percentage points of sales. On our performance improvement year-over-year, [here] has been primarily driven by fabricating \$9.9 million of subsea products within our Energy Industrial segment in Mexico versus in Rhode Island.

In Q3, company-level gross profit margins were up 23% and our gross profit of \$13.8 million is a \$20.2 million improvement over our gross loss of \$6.4 million during the same quarter last year. Our Energy Industrial segment delivered \$5.8 million of gross profit or 182% year-over-year increase. In EV Thermal Barriers, we delivered \$8 million of gross profit in Q3. If we compare this quarter with Q2, our EV thermal barrier gross profit improved by \$9.1 million, an incremental revenue of \$20.8 million.

Our third quarter of 2023 gross profit in EV Thermal Barriers was \$16.4 million higher than the gross loss of \$8.4 million that we incurred during Q3 of last year in this segment, reflecting the benefits of starting to operate at a revenue run rate that aligns with the size of our operation. The resulting gross profit margins during the quarter were 21% and 24% for [our] Energy Industrial and EV Thermal Barrier segments, respectively. Year-to-date, our gross profit of \$27.3 million reflects a \$36.7 million improvement in gross profit versus our gross loss of \$9.4 million during the same period last year.

Operating expenses which are sized for near-term projected annual revenue capacity of over \$550 million were \$28.4 million. This went up in the quarter above our target range as we had several one-time expenses, related to the reallocation of our resources from support functions and overhead to further building our technical capabilities. Approximately 1/3 of our quarter-over-quarter OPEX increase of \$3 million was driven by strategic investments in resources tied specifically to accelerating EV Thermal Barrier product development, and commercial launch activity with specific customers tied to unannounced OEM awards.

Putting these elements together, our adjusted EBITDA was negative \$7.3 million in Q3 compared to negative \$23.2 million during the same period last year, resulting in a \$16 million year-over-year reduction in our EBITDA loss, or a reduction of 68.5%. As a reminder, we define adjusted EBITDA as net income or loss before interest, taxes, depreciation, amortization, stock-based compensation, expenses and other items that we do not believe are indicative of our core operating performance.

In Q3, these other items included \$2.8 million of stock-based compensation, \$1.3 million of interest income, and \$2.2 million of employee retention credits from the government. Our net loss in Q3 decreased to \$13.1 million or \$0.19 per share versus a net loss of \$29.6 million or \$0.75 per share in the same quarter of 2022. Our quarter-over-quarter net loss decreased by \$2.3 million from \$15.4 million. Our year-to-date net loss of \$45.3 million is \$27.8 million lower than our loss of \$73.1 million during the first 3 quarters of last year or down by 38%.

Next, I'll turn to cash flow and our balance sheet. Cash used in operations of \$7.5 million reflected our adjusted EBITDA of negative \$7.3 million and cash used for working capital of \$1.5 million, offset by interest income of \$1.3 million. The key items that resulted in a usage of working capital were an increase in accounts receivable of \$10.2 million and inventory of \$1.1 million, offset by an increase in accounts payable of \$1 million and accrued expenses of \$6 million and a decrease in prepaid expenses of \$1.9 million.

Our capital expenditures during the quarter were \$32.3 million. These put our operating cash needs for the quarter at \$39.8 million. \$10 million of our CapEx was spent in closing the main buildings of Plant 2 in Georgia, and helping bring the plant to a healthy resting spot while securing the site. While the remaining \$22 million was spent on continuing to tool-up our facilities in Mexico to support the EV Thermal Barrier capacity ramp of 2024 and 2025, while covering the last invoices of the construction of our recently opened Advanced Thermal Barrier Center. The conversion of our remaining lines in Rhode Island to making PyroThin is also covered by these \$22 million.

We have incurred \$255 million in capital expenses through the end of the third quarter towards Plant 2 in Georgia. We ended the quarter with \$94.6 million of cash and shareholders' equity of \$409.8 million.

This balance is over \$20 million higher than our expectations going into the quarter, as we have managed to reduce the quarter-over-quarter cash burn of the company by \$33.9 million or 46% to \$39.8 million versus \$73.7 million in the prior quarter.

Turning over to Slide 5. I'd like to spend some time recapping the last 15 months and cover where we've been, before going into our updated financial outlook for the remainder of 2023. On the left side, you can see how we've improved the company's gross profit margins from 11% in Q1 of this year to 17% in Q2 and 23% during the most recent quarter. Our adjusted EBITDA loss has also shrunk from a loss of \$23 million in Q3 of 2022 to an adjusted EBITDA loss of \$7 million in the most recent quarter.

In Q3, it's no surprise that the lion's share of our 26% quarter-over-quarter revenue ramp was driven by 160% increase in demand for PyroThin, EV Thermal Barriers, for General Motors Ultium platform vehicles. GM has more than doubled its production run rates for Ultium powered vehicles from Q2 into Q3, and barring any interruptions, we expect this ramp to continue. This ramp in demand from General Motors drives the revenue guidance range to about \$225 million for 2023.

Running at an implied \$70 million or above quarterly revenue run rate for Q4 enables us to lower our EBITDA range to a loss of \$30 million to \$40 million, a loss reduction of \$15 million versus our prior range of negative \$40 million to negative \$55 million of adjusted EBITDA for the year.

As we factor in the effect of meaningful interest income and the different amortization schedule as we operate with less deployed capital, we're also lowering our net loss guidance for the year from a loss of \$75 million to \$85 million, to a loss of \$52 million to \$62 million. This improvement of \$17 million represents an 18% and 17% reduction on the lower and upper end of our prior guidance range, respectively. This also brings our earnings per share guidance to an updated loss range of \$0.76 per share to \$0.90 per share.

With \$147.7 million of CapEx spent year-to-date, we realized that we're very close to our previously stated CapEx guidance of \$150 million for the year. Last quarter, we mentioned that we would only increase this amount if we saw a very clear picture of a 2024 EV Thermal Barriers demand increase as the second half of 2023 materialize.

Now that we've experienced this ramp, we think that it is prudent to spend up to an additional \$25 million in Q4 of 2023, to enable additional capacity improvements for Thermal Barrier assembly in Mexico and improvements to our aerogels plant in Rhode Island as well as preserving the value of our assets in Georgia.

For Plant 2 in Georgia, October was really the first month at which the spend level crossed below our target level to preserve the site now that it has arrived at a healthy resting spot with all the equipment that is on site, secured in a temperature-controlled environment. These \$25 million would bring our CapEx spend for 2023 to \$175 million, and enable us to flex up to supply a plan that isn't as constrained as our original expectations for 2024.

In the near term, we continue to focus on managing the company with at least \$75 million of cash on the balance sheet, and are pursuing nondilutive sources of financing such as working capital, lines of credit, asset-backed loans, equipment leases and other instruments that leverage our current asset base. You may remember that on June 15, we terminated our ATM program and that we have not sold any equity in 2023.

Turning over to Slide 6. At this point, you've heard Don and I outline for several quarters and in every meeting that our team is focused on making the most of our existing assets and leveraging our partner in China to have approximately \$550 million of revenue capacity, and delivered 35% gross margins as we get close to maximizing this capacity.

While I'll let you digest this slide on your own, showing how our quarterly results of 2023 compared with this North Star, it's easy to see how on a run rate basis, the profitability of our business has been positively evolving during the first 3 quarters, with progressive increases in gross profit from 11% in Q1, 17% in Q2, and most recently, 23% in Q3 at a \$244 million annual revenue run rate that is still only 44% of our potential capacity.

When we compare the last 12 months to the prior 12 months, our revenue run rate has increased by 41%, while our material costs have decreased by 17 percentage points of sales, and our manufacturing costs have decreased by 10 percentage points of sales. Our OPEX has basically remained flat as a percentage of sales, and these have enabled us to shrink our EBITDA loss by 47%. These results are the benefit of rightsizing and right timing our capacity ramp.

And as we look ahead to '24 and 2025, we will continue adopting the same approach to posting results on the board without being influenced by headlines of a slowing demand for EVs and retimed OEM investments. We believe that we may have been very conservative and cautious in our plans and are encouraged to see the updated reality continue to be above our expectations.

The OEM ramp delays of the past 18 months and rising cost of capital have taught us a lot, and we are confident that these learnings will enable us to continue playing offense by posting results, and then making investments versus the treadmill of constantly playing defense by working the other way around.

With that, I'm happy to turn the call back to Don.

Donald R. Young
President, CEO & Director

Thank you, Ricardo. We have covered a significant amount of ground today in reviewing Q3 and our near-term outlook. Before we move to Q&A, I would like to emphasize our focus on driving significant profitability from our existing resources and opportunities. We believe the near-term business profile as we have it constructed, and again, consistent with our current assets and commercial opportunities, has the potential to produce on an annual basis approximately \$550 million of revenue, approximately \$200 million of gross profit, and approximately \$140 million of EBITDA.

We believe Q4 will be the next significant step towards this level of business performance. At the same time, we believe that we maintain our full longer-term upside potential as we continue to win design awards from EV OEMs, to expand our profitable baseload of Energy Industrial revenue and to leverage our aerogel technology platform into additional high-value markets, including our ongoing work in battery materials. The key point is that we are seeking to avoid unnecessary dilution for our shareholders by optimizing the use of our existing assets and opportunities to create a dynamic and cash-generating business.

With that, we welcome the Q&A session.

Question and Answer

Operator

[Operator Instructions] Our first question today comes from Colin Rusch with Oppenheimer.

Colin William Rusch

Oppenheimer & Co. Inc., Research Division

With the cadence of these incremental wins and the start of production, can you talk a little bit about the dynamics around selling to just fill the channel and [get] -- if you guys supplied up to start production, and how much actual sell-through you're seeing on the vehicles at this point?

Ricardo C. Rodriguez

CFO & Treasurer

Sorry, Colin, maybe just to clarify, so sell-through on the end of General Motors specifically? Or is this more related to additional OEMs?

Colin William Rusch

Oppenheimer & Co. Inc., Research Division

The dynamic we're trying to get after is how much of the growth is really coming from selling into the OEMs? And as you have an incremental cadence of new program wins, there's going to be growth to support that production. But there's a lot of concern around how these vehicles are selling, and potential stagnation in terms of growth at some point? So that's the dynamic we're trying to get after.

Ricardo C. Rodriguez

CFO & Treasurer

Yes. I mean I think there's sort of 2 parts to thinking about this one, right? I mean we've known that some of these EVs have been sitting in lots for over a year at this point, right? And we factored that into our conservatism as we factor in the volume plans, not only when we are making our own plans, but also when we're quoting these opportunities.

And when it comes to Ultium specifically, it's a little tough to assess how close General Motors is to hitting the demand limit, when they only produce -- we estimate around 14,000 vehicles in Q3. And there's roughly 3,000 to 4,000 Chevy dealers in North America and that doesn't even come close to getting up to the limits of the consumer demand for these vehicles. So, we do think that GM has ways to go in terms of ramping up production before they start to see any limit on consumer demand.

Now as we look at additional opportunities, going to the second part of the question, I mean we -- in the same way as we planned the ramp from GM, and we truly have learned a lot through the delays with GM's ramp on how to deploy capital in these opportunities, how to design the processes to execute them. And we apply a very similar level of discounting to some of these opportunities in 2025 and 2026 that we're being awarded here in the near term.

And it's worth noting that once an OEM puts together a sourcing package for a vehicle that is 2 to 3 years away, a lot of work has already been done and a lot of investment has already been made. And so, we do feel pretty good about our ability to get more wins for this 2025 and 2026 time-frame.

Now additional programs, I think, for that 2027 and 2028 time-frame, we do expect that to continue being retimed, and frankly, see the recent news of -- some retime BV investments is something that was kind of inevitable just given everyone's cost of capital. And we were surprised that it actually took this long to ratchet back some of these investment expectations. But we still feel very good about the opportunities that we have in the horizon for 2025, 2026. And even if we go back to our plans in 2021, we have not had to adjust those downward based on what we're seeing.

Donald R. Young

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President, CEO & Director

Colin, I would just add, I think the momentum that we've had moving from Q2 to Q3, I think you'll see that momentum continue for us here as we move into Q4 and into the new year.

Colin William Rusch

Oppenheimer & Co. Inc., Research Division

And then as you [were] contracting on new product wins with these new customers, the value, both mechanical and the thermal performance of the material is becoming really well understood. I guess I'm curious about any potential for price leverage on your part, or at least price preservation even as you get into higher volumes with some of these customers given that mechanical performance?

Ricardo C. Rodriguez

CFO & Treasurer

Yes, I mean, for us, it's really not just about the underlying merits of the material and the critical problems that it solves, but also the amount of capital that we've deployed and that we will continue to cautiously deploy here and an emphasis on paying it back, right? So, we will manage pricing in a way that kind of come hell or high water, we really align with the gearing that we've presented here on Slide 6, for example, right? I mean, we want all programs and all sales that we execute here to not pull us away from this path of getting 35% gross profit margins, and we'll manage pricing to do that.

Operator

We now turn to Eric Stine with Craig-Hallum.

Eric Stine

Craig-Hallum Capital Group LLC, Research Division

So maybe for me, I know in the past, you had talked about that \$550 million run rate, 25% EBITDA margin, and very good slide, appreciate that, that you gave this morning. I think previously, you'd said -- thought that was possible in the next 4 to 6 quarters. And I know you're managing the contract manufacturer and ramping things up and you're constrained in EI and all of that. Is that still something that you see as possible on a run rate basis by, I guess, your expectation had been 3Q of '24?

Ricardo C. Rodriguez

CFO & Treasurer

It's not unlikely. I mean, if -- we actually included a slide in the back for everyone's benefit on -- if you just look at GM alone and the ramp that they're in, right? So, kind of in spite of the headlines, if you just look at the IHS ramp and if you see GM's intended ramp, it's even higher than this. But there's a potential for them to just continue in doubling production every quarter for at least 2 more quarters, right? And the implications of that on our revenue run rate supplemented by the supply from China for our Energy Industrial business put us on this path to be able to get there. Now whether it happens in Q2 or Q3, Q4, we're just cautiously managing it, but we do see it as something that is still possible.

Eric Stine

Craig-Hallum Capital Group LLC, Research Division

Yes, just making sure that's still kind of the expectation. And then maybe for my follow-up, maybe just an update on the DOE loan that you're in the mix for. I know previously, you talked about potentially hearing kind of next steps by year-end, and the expectation that if you're invited for further diligence that, that's a great sign towards the eventual award or possible award. Just curious, if that's still the view?

Donald R. Young

President, CEO & Director

The DOE -- the LPO process -- we continue to be closely engaged with the LPO office and the team, if you will, on that side of the table and working very closely with them. Our team is doing a great job. I think we're putting our best foot forward. And of course, it's always an unpredictable process. But we do believe

that we're on that time line that we outlined earlier, and we hope to be able to keep you posted on that here as we work our way through Q4.

Operator

We now turn to Alex Potter with Piper Sandler.

Alexander Eugene Potter

Piper Sandler & Co., Research Division

So, I was wondering just maybe -- obviously, you're working on the contract manufacturing avenue with your partner in China. Any update you can give on how that's progressing? What gives you confidence that everything is on track and on time?

Donald R. Young

President, CEO & Director

Well, we -- again, that's -- we're impressed by them. We're working very closely. It's a matter of qualifying our products, setting up the supply chain, the logistics, working with our Energy Industrial customers. And it's going very well. And we have said that we will test the system here over the course of 2023. And we're doing just that now and that -- it should be up and contributing at the early part of 2024. And that's where -- we're right on that path. So, we're confident, we're impressed by the work that we're doing together with our supplemental supplier. And again, right on track.

Alexander Eugene Potter

Piper Sandler & Co., Research Division

And then the second question I had was on the PyroThin volume that you're selling into GM for the Ultium products. Just kind of curious, are they putting PyroThin themselves in their own inventory? Or are they taking it from you, putting it directly into a vehicle and pushing the vehicle out? I'm just trying to understand -- obviously, you're producing, you sell to them, but I don't know if -- just, I guess, trying to gauge whether there's the potential for inventory buildup or shortages at any point between you, GM and then downstream of GM?

Ricardo C. Rodriguez

CFO & Treasurer

Yes. I mean, I think the best way to think about it is that -- it's fair to say that a part that we're making in Mexico right now probably won't find its way into a finished vehicle for about 1 month. And so, when the ramp is as significant as what we're seeing, right, looking at the 14,000 vehicles -- Ultium-based vehicles that we believe GM built in Q3, isn't totally enough to explain our revenues, right? You'd -- if you take our revenue divided by 14,000, you end up with a CPV that's way higher than what's actually there. And so, GM is building up a decent amount of inventory here to enable the ramp.

And from what we understand, I mean, they've been hand to mouth here during Q3. And we'll have to see really -- kind of, going back to Colin's question on, we'll see how long these vehicles then spend on the lot to translate that into ultimate sales for GM. But we do see about a 1-month delay, which kind of gives us the chance to recognize revenues and build revenues of 1 month ahead of the end of the quarter, right?

Operator

Our next question comes from Jeff Osborne with TD Cowen.

Jeffrey David Osborne

TD Cowen, Research Division

Just 2 quick ones on my side. I was curious on the Audi win, that's great to see. Is that part of a broader program as part of the VW family, that maybe it's just starting with one vehicle in '24? Any help there would be helpful because I thought you were targeting more [pack] designs across multiple vehicles as opposed to a one-off. So, understanding that would be helpful.

And then, any other updates that you could share on the battery materials side of the business? You haven't talked about that in a few quarters, but I'm just curious how that's progressing? Would be also helpful to understand.

Ricardo C. Rodriguez

CFO & Treasurer

Yes. So happy to jump in on the first one there, Jeff. Yes, so, the Audi program is an electric vehicle platform, which means that multiple different Audi nameplates could be based on this platform.

Donald R. Young

President, CEO & Director

And then on the battery materials side, we continue to work on advancing the technology. And we are endeavoring to solve a hard problem, and we are making, I think, steady and interesting progress on that. We've kept quiet on it to a great degree on purpose as we've continued to advance that. I think it will be an opportunity for us to talk about it in the coming earnings call or 2. And I would really like for that to not only talk about the status of the technology, but perhaps also the status of 1 or 2 working relationships we have with third-parties at that point, helping us advance the technology and taking that next step. But it's a very exciting program for us. We've got an excellent team of people working on it, and it's -- we're careful in our expenditures, but it's adequately and well-resourced to continue to advance that technology.

Operator

We now turn to George Gianarikas with Canaccord.

George Gianarikas

Canaccord Genuity Corp., Research Division

Maybe just to focus on some of the potential EV OEM awards that you haven't announced yet, that could get announced by the end of the year. Curious, can you just shed a little light on when you expect potential volumes to start there just so we can understand? And when we look to 2024, how diversified will your PyroThin business be away from GM?

Donald R. Young

President, CEO & Director

Yes. So, I mean, those awards would really kick in, in 2025. The meaningful volumes and the ramp would happen in 2025. So, for 2024, it's fair to say that we will still be fairly concentrated on GM, which is proving out to not be a bad thing now. And then Toyota and the Scania will also ramp up in the second half of next year.

George Gianarikas

Canaccord Genuity Corp., Research Division

And maybe to the extent you could share any details on Toyota? I mean, they seem to be -- at least in their public pronouncements going back and forth with their commitment to EVs. I mean what sort of momentum do you see with that account?

Ricardo C. Rodriguez

CFO & Treasurer

Yes. I mean, I think we would put them in a similar bucket as the other OEMs. I mean they still haven't announced a broad battery platform strategy, instead it's more of a nameplate-by-nameplate approach. And that scope can increase, but we don't see anything here in the near term for '24.

Operator

We now turn to Chris Souther with B. Riley.

Christopher Curran Souther

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B. Riley Securities, Inc., Research Division

Could you talk through the moving pieces on the revenue and EBITDA guidance for the fourth quarter? You said revenue is expected to grow. Gross margins are expected to increase, and you have a \$7 million EBITDA loss in the third quarter. So, what are the moving pieces around the low end of the EBITDA guidance, yes, \$8 million loss? Can you talk through what you'd need from a revenue perspective to hit the positive EBITDA in the fourth quarter? Or is it some specific output out of China or other factors that would kind of be in play there?

Ricardo C. Rodriguez

CFO & Treasurer

Yes. No, that's a good question. We knew you'd kind of go there. And we really are protecting for the doomsday scenario here. If you look at the negative \$40 million of EBITDA, right -- of potential EBITDA, I think we'll be forthcoming tightening that up as November materializes here. But for us, really, when we looked at the UAW agreement not being ratified, some of the big certification work streams with the contract manufacturers still in full swing, and the potential cost of an additional turn of certification on those products. Really, if you combine all of those negative things that could happen, albeit with low probabilities, that's how you end up at a negative \$40 million of EBITDA. And yes, I mean, I do think that given the ramp that we expect for Q4, we will have an opportunity to potentially tighten that range.

Donald R. Young

President, CEO & Director

Yes, Chris, I would say that there's a reasonable chance that we'll provide one more update before year-end on some of these topics that you and your colleagues have asked about, revenue and EBITDA, additional OEM awards, commentary on the DOE LPO, those sorts of things. I think it could very well be appropriate for us to -- again, to update you all one more time before year-end.

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

And then, just -- you kind of talked through your sales not matching up, kind of as a leading indicator of GM production, because obviously, the implied ASPs otherwise would have been very high here. Where are the ASPs kind of shaken out, if we exclude the Scania stuff, just for kind of general cars, SUVs that are in your OEM awards? And how much revenue are you guys getting from awards that you haven't yet received as far as kind of component sales that are related to testing and the like? I just wanted to see if you could kind of provide a bit more clarity around that.

Ricardo C. Rodriguez

CFO & Treasurer

Yes. I mean the prototype revenues right now make up, well, less than 10% of our PyroThin revenues. And then I'm assuming you mean more around -- in -- when it comes to the rest of the PyroThin revenues, you mean more [around] our content per vehicle, right, how that is tracking?

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

Yes. Exactly.

Ricardo C. Rodriguez

CFO & Treasurer

Yes. And so, I mean, we're still pretty consistently seeing our content per vehicle in that \$700 to \$1,000 per vehicle range given that the bulk of what we're supplying is vehicles with pretty large packs and pouch cells, right? And when we add one of these prismatic programs, I think the content there is more in the \$350 to \$400 per car range. And -- but nonetheless, I mean, it's a much simpler part to produce. It requires a lot less capital. And so, we're happy with the economics of those awards as well. But right now,

I mean, if you're looking at what we're producing here -- what we produce here in Q3, what we'll do in Q4 -- and most of '24, it's still going to be this large pouch configuration with the CPV in that \$700 range.

Operator

Our next question comes from Tom Curran with Seaport Research Partners.

Thomas Patrick Curran

Seaport Research Partners

[So] when it comes to the supplemental supply that you're in the process of testing with your Chinese contract manufacturing partner, is the expectation still that once that's fully ramped to the maximum expected capacity that your available quarterly revenue capacity for Energy Industrial would be around \$37.5 million? And then, when it comes to that \$118 million backlog of unfilled orders, how much time do you feel you have to catch up with those orders? And do you have any risk or liability related to letting them remain unfilled past a certain point in time?

Donald R. Young

President, CEO & Director

Well, it's a good question, Tom. We are very close to our Energy Industrial customers and the distribution channels and the engineering firms, and we actively communicate with them to be sure that we're meeting their expectations. And -- but we are in a position to switch over to the supplemental supply, again, early next year. And we feel that we will fulfill our obligations, our responsibilities to those customers. And quite frankly, that business is strong.

Our sales team on the Energy Industrial side is very active on both, the Pyrogel side of the business and the Cryogel side, at the LNG part of the business. And so that business is going to continue to grow and be an important baseload of revenue and gross profit for our company. And so again, we feel we're in a good position with those customers communicating well. Yes, the backlog -- and you might remember if you go back to the last time, we were short of capacity. The backlog is substantial. And -- but we're in good shape. So long as we communicate well, I think we're good.

Ricardo C. Rodriguez

CFO & Treasurer

Yes. And if I may add, I mean, the team is pretty good at assessing what gets into that \$118 million backlog and which orders they accept. And an order makes it into that \$118 million if the theme has line of sight to fulfilling within a reasonable time frame.

Thomas Patrick Curran

Seaport Research Partners

That's reassuring. And just to follow up and try to clarify, within the \$550 million, would you think of fully -- eventual fully available annual revenue capacity for EI being around \$150 million?

Donald R. Young

President, CEO & Director

Yes, we would. I mean when we think about \$550 million, we do think about the PyroThin part of that being [\$400 million] and the remaining part being the Energy Industrial side of our business.

Thomas Patrick Curran

Seaport Research Partners

And then turning to EVTb, when it comes to these next 2 potential awards that you remain very optimistic about landing before year-end, would you still expect at least one of those to result in a new customer, essentially what would be customer #4?

Ricardo C. Rodriguez

CFO & Treasurer

Yes, that's correct. Yes. I mean we kind of put your bingo board there on Slide 3 and we would expect different logos on there.

Thomas Patrick Curran
Seaport Research Partners

Apparently you know I love playing my bingo, I'm almost 50.

Operator

This concludes our Q&A. I'll now hand back to Don Young, CEO, for closing remarks.

Donald R. Young
President, CEO & Director

Thank you, Elliot. We appreciate your interest in Aspen Aerogels. We look forward to staying in close touch with you and reporting our fourth quarter 2023 results to you early next year. Be well, and have a good day. Thanks so much.

Operator

Ladies and gentlemen, today's call has now concluded. We'd like to thank you for your participation. You may now disconnect your lines.

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